Task 12 – Log Monitoring & Analysis

Tools Used

- Linux authentication logs
- Windows Event Viewer
- Splunk Free

1. Understand Log Types

I learned that logs store system activities.

In **Ubuntu**, security logs are inside `/var/log/auth.log`.
In **Windows**, login logs are stored in **Event Viewer** → **Security**.

This showed me that both systems record user activity but in different formats.

Ubuntu

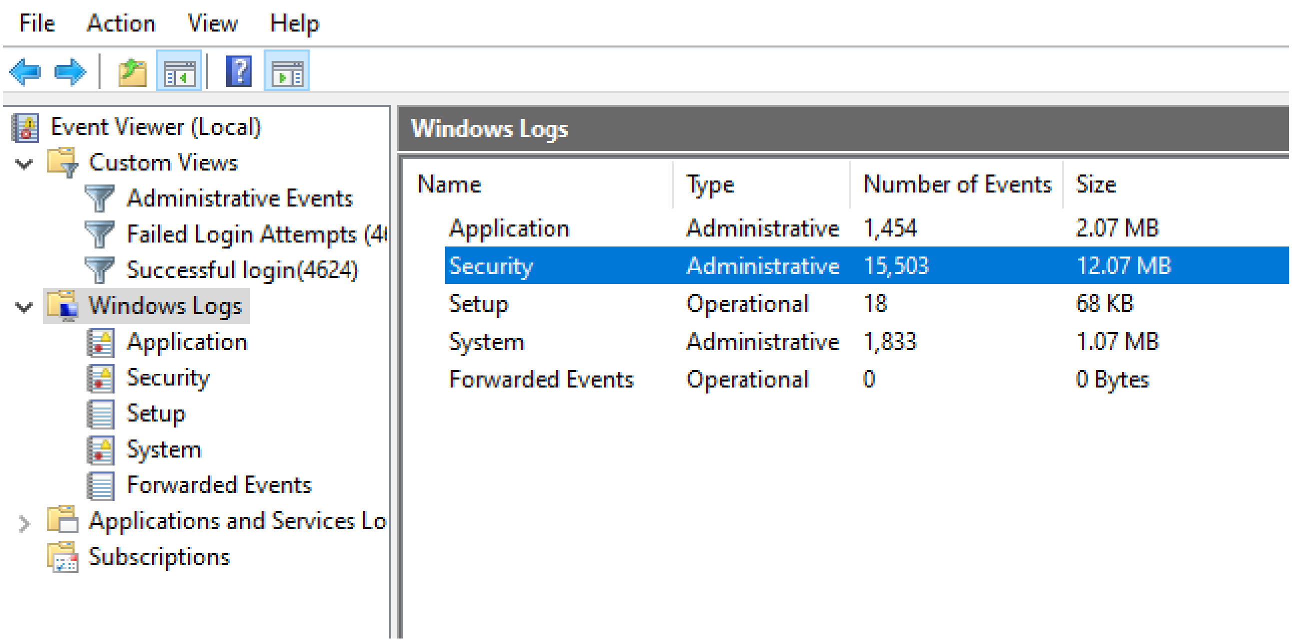
Screenshot : Linux `/var/log` directory

```
mounika@ubuntu:~$ ls /var/log
alternatives.log        dist-upgrade      lastlog
alternatives.log.1      dmesg             mysql
apache2                 dmesg.0           openvpn
apport.log              dmesg.1.gz        private
apport.log.1            dmesg.2.gz        README
apt                     dmesg.3.gz        speech-dispatcher
auth.log                dmesg.4.gz        sssd
auth.log.1              dpkg.log          syslog
auth.log.2.gz           dpkg.log.1        syslog.1
boot.log                faillog           syslog.2.gz
boot.log.1              fontconfig.log    sysstat
boot.log.2              gdm3              ubuntu-advantage-apt-hook.log
bootstrap.log           gpu-manager.log   ufw.log
btmp                    hp                ufw.log.1
btmp.1                  installer         unattended-upgrades
cloud-init.log          journal           vbox-setup.log
cloud-init-output.log   kern.log          wtmp
cups                    kern.log.1
cups-browsed            kern.log.2.gz
mounika@ubuntu:~$
```

`auth.log`, `syslog`

Windows VM :

Screenshot : Security log window



2. Analyze Authentication Logs

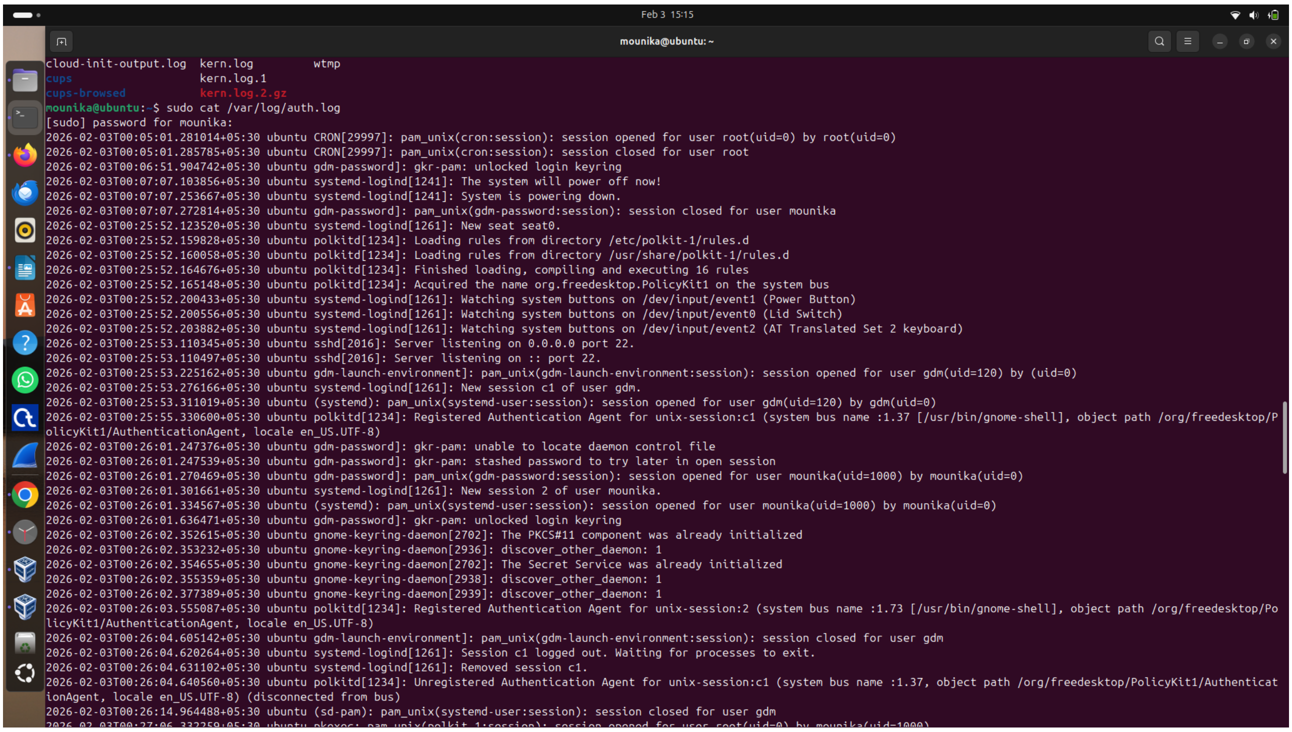
In Ubuntu, I opened `auth.log` to see all login and sudo activities.
In Windows VM, I checked Security logs using Event Viewer.

This helped me understand how user authentication is recorded in both systems.

Ubuntu

```
sudo cat /var/log/auth.log
```

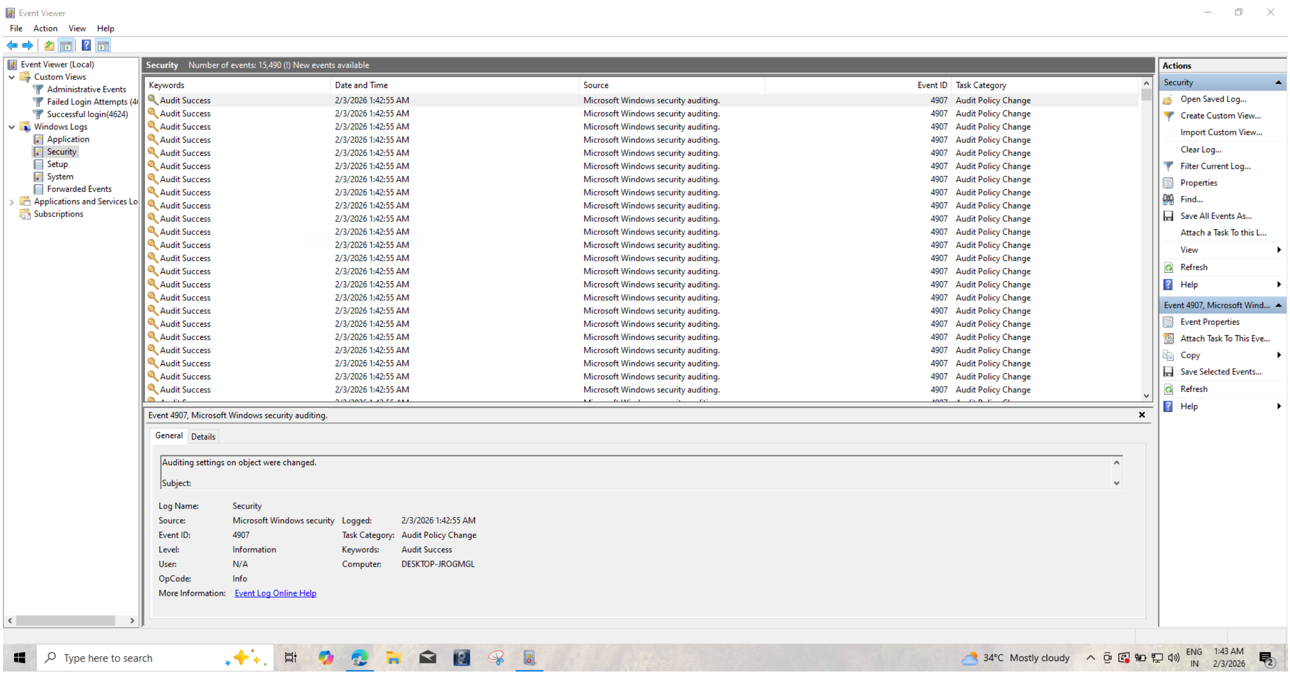
Screenshot : auth.log open



Windows

Event Viewer → Security → Scroll

Screenshot : Security log entries list



3. Identify Failed Logins

Using grep in Ubuntu, I filtered failed and successful logins.
In Windows, I filtered Event IDs **4625** and **4624**.

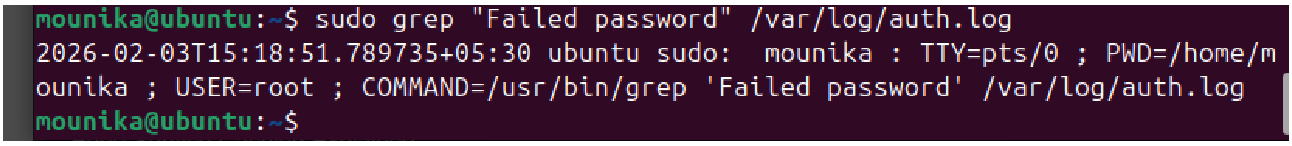
This helped me clearly separate attack attempts and real logins.

Ubuntu

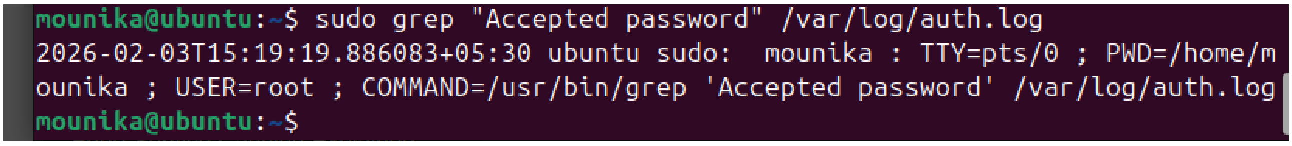
```
sudo grep "Failed password" /var/log/auth.log

sudo grep "Accepted password" /var/log/auth.log
```

Screenshot : Failed logins



Screenshot : Successful logins



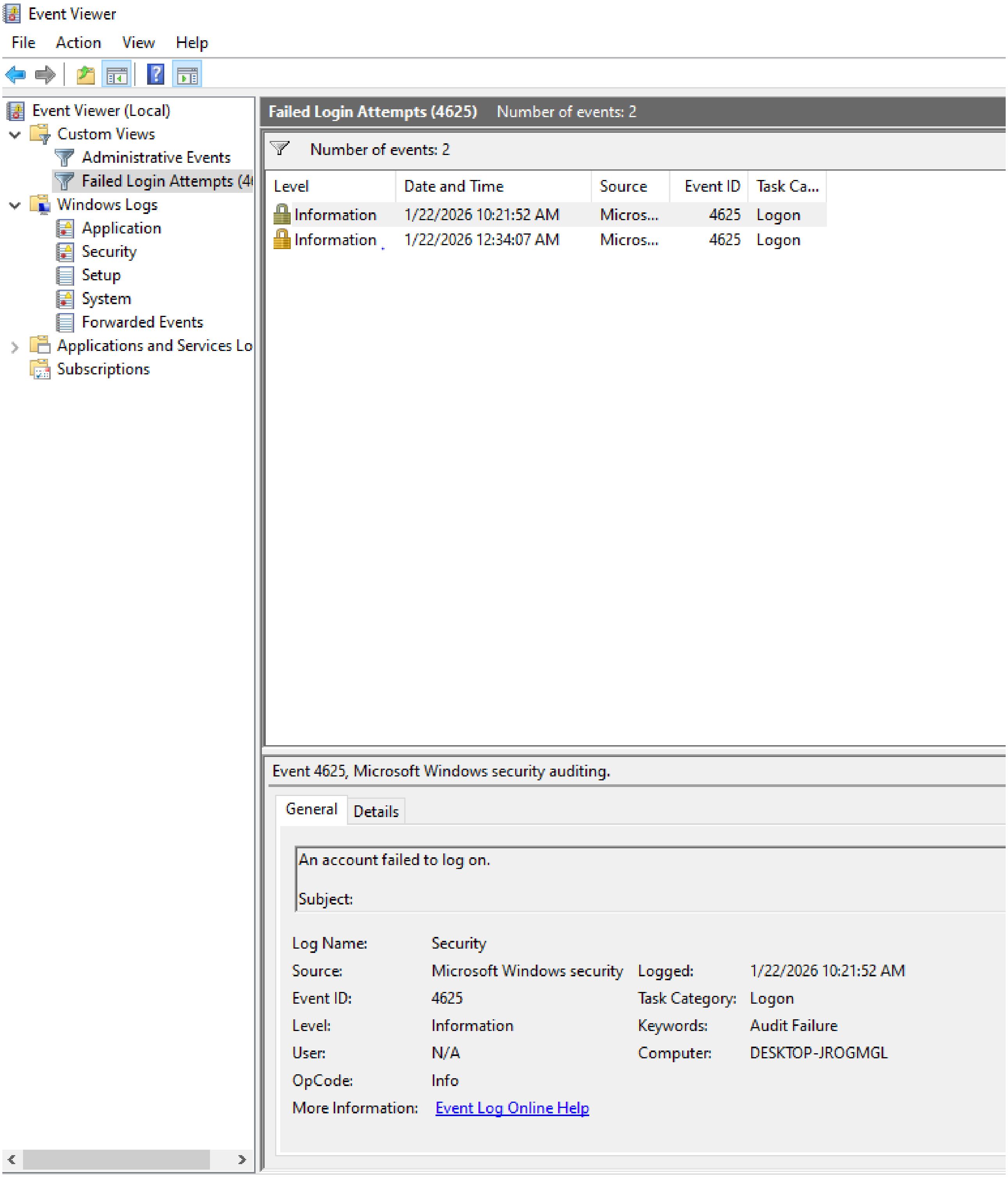
Windows

Filter logs:

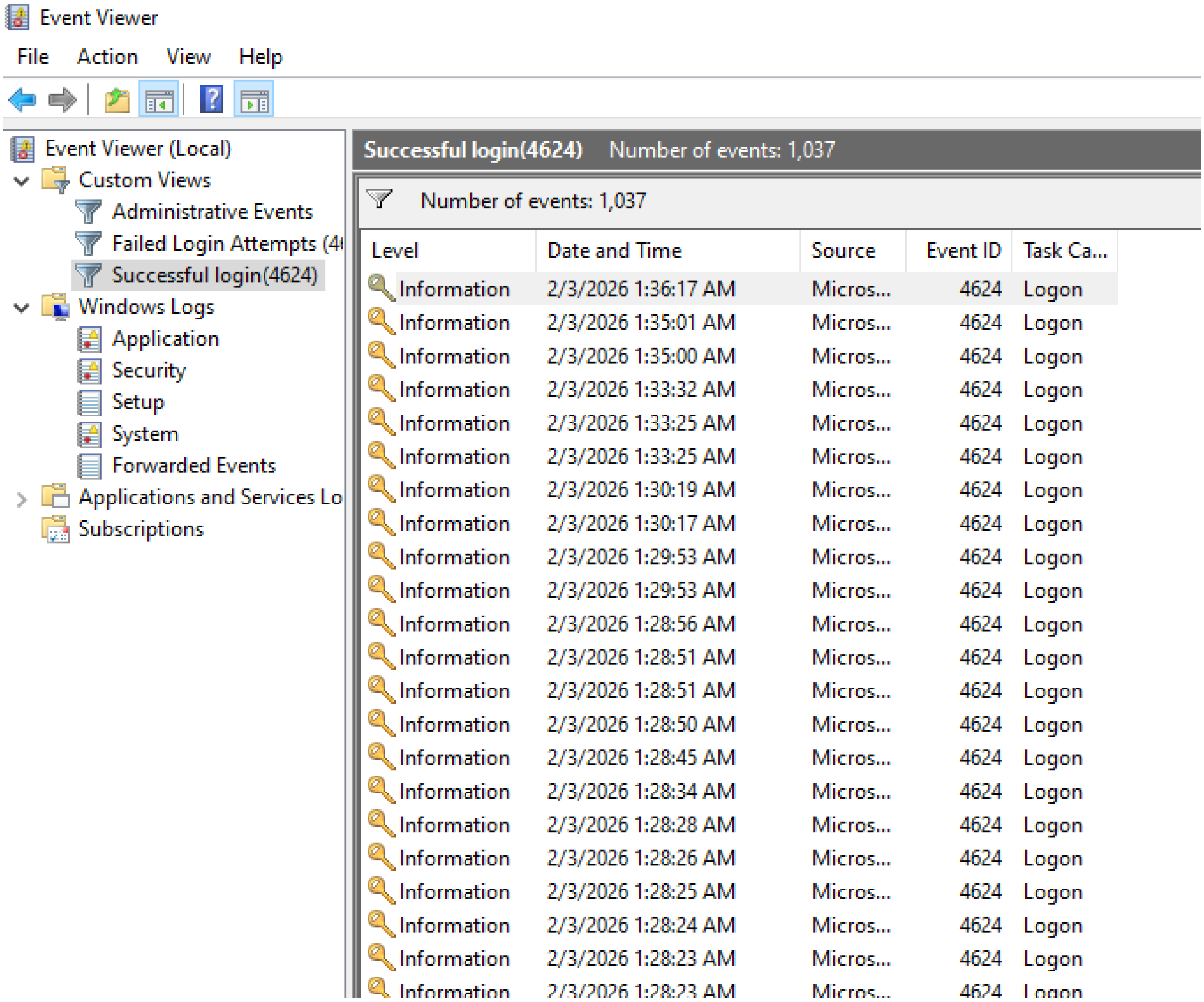
Right click **Security** → **Filter Current Log**

- Event ID: **4625** (failed)
- Event ID: **4624** (success)

Screenshot : Failed login event details



Screenshot : Successful login event details



4. Detect Anomalies

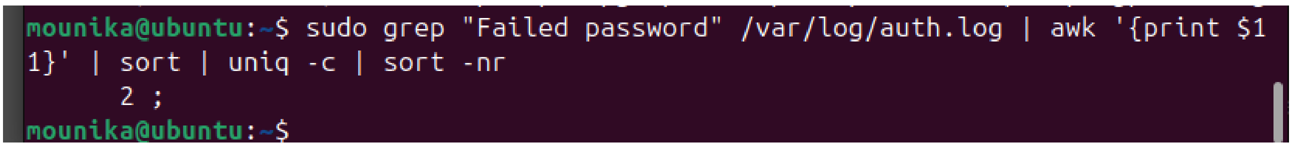
In Ubuntu, I found repeated login failures from the same IP.
In Windows, I saw many failed logins for the same user.

These patterns indicate possible brute force attacks.

Ubuntu

```
sudo grep "Failed password" /var/log/auth.log | awk
'{print $11}' | sort | uniq -c | sort -nr
```

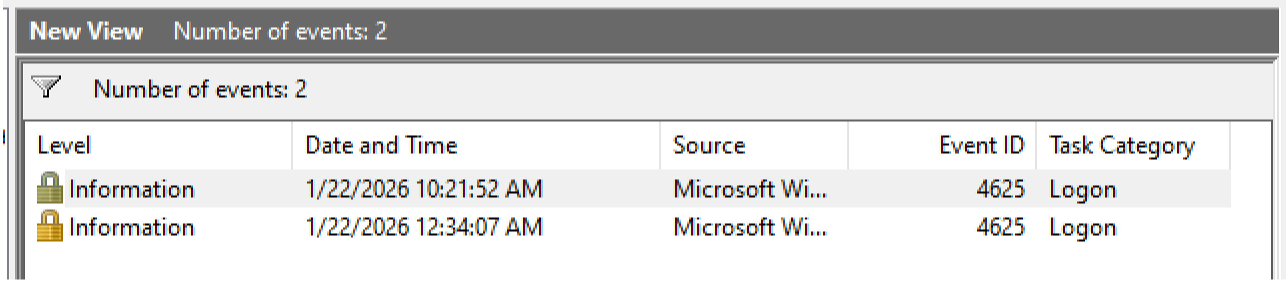
Screenshot : Repeated IP attempts



Windows

Look for many 4625 events from same user/time

Screenshot : Multiple failed logins



5. Correlate Events

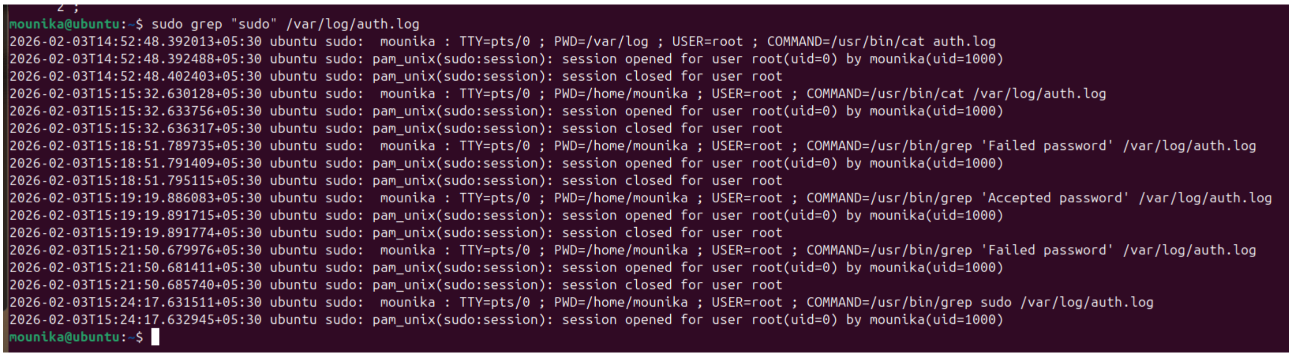
In Ubuntu, failed logins followed by sudo commands showed possible compromise. In Windows, Event ID **4672** showed privilege usage after login.

This correlation proves how attackers move after gaining access.

Ubuntu

```
sudo grep "sudo" /var/log/auth.log
```

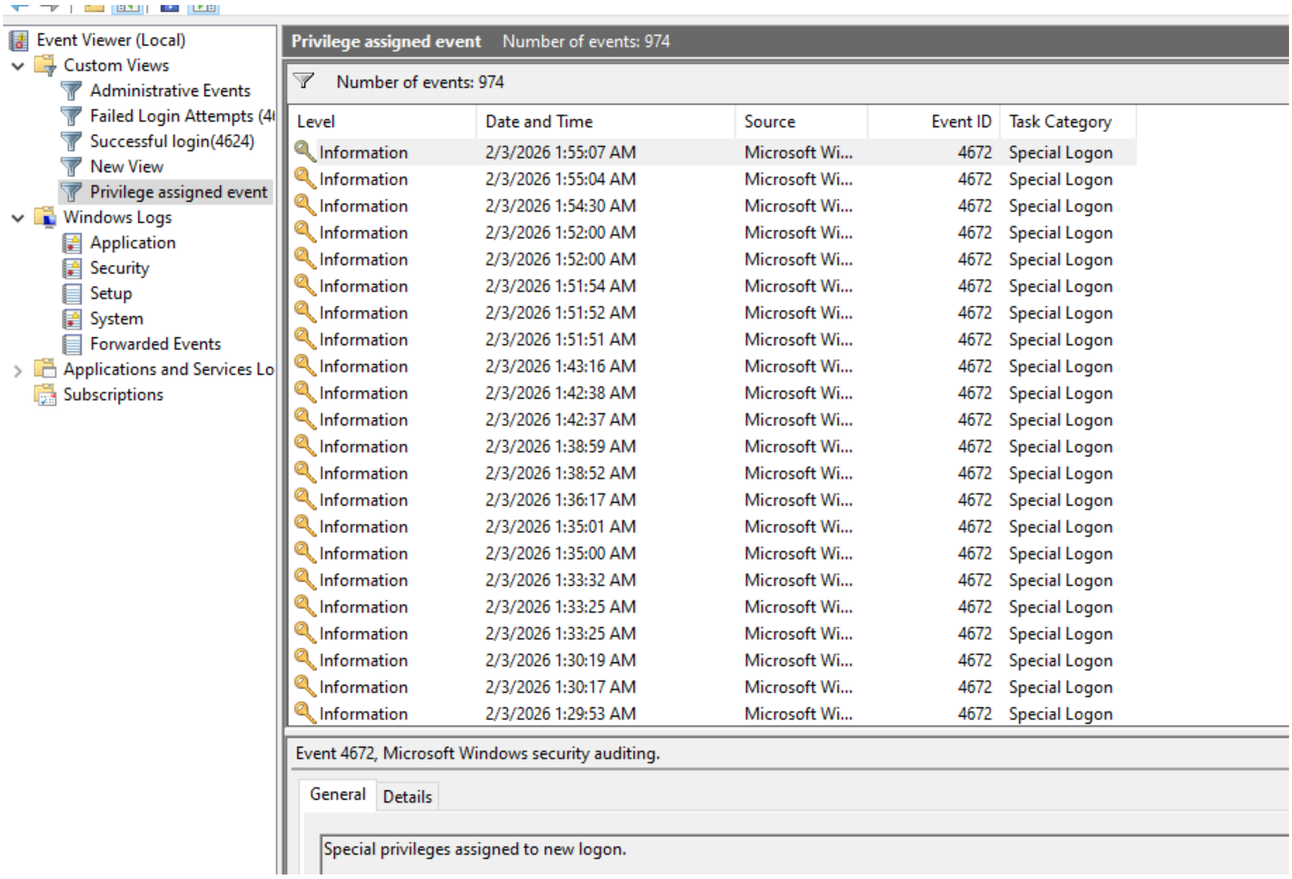
Screenshot : sudo activity



Windows

Filter Event ID **4672**

Screenshot : Privilege assigned event



6. Learn SIEM Basics

From this task, I learned that a **SIEM (Security Information and Event Management)** system is used to collect logs from multiple systems like Linux, Windows, firewalls, and applications into one centralized platform.

Instead of checking each system manually, SIEM tools automatically:

- Collect and normalize logs
- Detect suspicious patterns
- Correlate events
- Generate real-time alerts

Examples of SIEM tools include Splunk, IBM QRadar, and ArcSight.

Even though I did not install a SIEM tool, I understood how it would simplify monitoring by showing all security events in one dashboard and alerting administrators when threats occur.

7. Write Alerts

Trigger	Alert
>5 failed logins	Brute force
Admin login from new IP	Critical
Login at odd hours	Suspicious

8. Document Findings

By documenting logs from both Ubuntu and Windows, I learned how security teams track and stop attacks.

9.Conclusion

This task helped me understand real-world log monitoring and incident detection using both Linux and Windows systems.